

LOGIC AND PROOFS

Proofs

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OUTLINE

1. DIRECT PROOF
2. PROOF BY CONTRAPOSITION
3. PROOF BY CONTRADICTION
4. PROOF BY CASES
5. PROVING NON-CONDITIONAL STATEMENTS

- **Theorem:** A mathematical statement that can be shown to be true.
 - Theorem is usually reserved for a statement that is considered at least somewhat important.
 - Less important theorems sometimes are called propositions.
- **Axiom (or postulate):** A statement that is assumed to be true.
- **Lemma:** A less important theorem that is helpful in the proof of other results.
- **Corollary:** A theorem that can be established directly from a theorem that has been proved.
- **Conjecture:** A statement that is being proposed to be a true statement, usually on the basis of some partial evidence.
- **Proof:** A valid argument that establishes the truth of a Theorem.
 - The statements used in a proof can include axioms, definitions, the premises, if any, of the theorem, and previously proven theorems and uses rules of inference to draw conclusions.

Definition

- The integer n is even if there exists an integer k such that $n = 2k$.
- The integer is odd if there exists an integer k such that $n = 2k + 1$.
- Two integers have the **same parity** when both are even or both are odd; they have **opposite parity** when one is even and the other is odd.

Direct Proof

A direct proof of a conditional statement $p \rightarrow q$ is constructed when the first step is the assumption that the antecedent p is true.

Subsequent steps are constructed using rules of inference, axioms, definitions and/or previously proven theorems with the final step showing that the consequent q must also be true.

Outline for Direct Proof.

Proposition If P , then Q .

Proof. Suppose P .

$\vdots$

Therefore Q . ■

Example

Give a direct proof of the theorem “If n is an odd integer, then n^2 is odd.”

Solution

This theorem states that $\forall n P(n) \rightarrow Q(n)$ where
 $P(n)$: “ n is an odd integer”
 $Q(n)$: “ n^2 is odd”

Assume that the hypothesis of this conditional statement is true, i.e. we assume n is odd.

$\therefore n = 2k + 1$, where k is some integer $\leftarrow$ by definition of an odd integer.

We want to show that n^2 is also odd. Squaring both sides of $n = 2k + 1$, we obtain

$$\begin{aligned} n^2 &= (2k + 1)^2 \\ &= 4k^2 + 4k + 1 \\ &= 2(2k^2 + 2k) + 1 \\ &= 2l + 1 \quad \text{where } l = 2k^2 + 2k, \text{ and } l \in \mathbb{Z} \end{aligned}$$

We can conclude that n^2 is an odd integer.

Consequently, we have proved that if n is an odd integer, then n^2 is an odd integer.

Example

Prove that the sum of two even numbers is even.

Solution

This could be rephrased as:

If x, y are even, then $x + y$ is even.

Assume x and y are even.

$x = 2m, m \in \mathbb{Z}$ by definition of even number

$y = 2n, n \in \mathbb{Z}$ by definition of even number

$$\begin{aligned}x + y &= 2m + 2n \\&= 2(m + n) \\&= 2r \quad \text{where } r = m + n, \text{ and } r \in \mathbb{Z}\end{aligned}$$

That is, $x + y = 2r$ is an even number. We have proved that the sum of two even numbers is even.

Proof by Contraposition

A type of **indirect proof**.

Makes use of the fact that the conditional statement $p \rightarrow q$ is equivalent to its contrapositive, $\neg q \rightarrow \neg p$.

We take $\neg q$ as a premise, and show that $\neg p$ must follow.

Outline for Contrapositive Proof.

Proposition If P , then Q .

Proof. Suppose $\neg Q$.

$\vdots$

Therefore $\neg P$. ■

Example

Prove that if n is an integer and $3n + 2$ is odd, then n is odd.

Solution

Assume that n is even, i.e. $n = 2k$, for some integer k , by definition of an even integer. Substituting $2k$ for n , we find that

$$\begin{aligned} 3n + 2 &= 3(2k) + 2 \\ &= 6k + 2 \\ &= 2(3k + 1) \\ &= 2r \quad \text{where } r = 3k + 1, \text{ and } r \in \mathbb{Z} \end{aligned}$$

This tells us that $3n + 2$ is even. This is the negation of the premise of the theorem. Because the negation of the conclusion of the conditional statement implies that the hypothesis is false, the original conditional statement is true.
 $\therefore$ We have proved that “If $3n + 2$ is odd, then n is odd.”

Practice

Prove that if n^2 is odd, then n is odd.

Definition

The real number r is rational if there exist integers p and q with $q \neq 0$ such that $r = p/q$.
A real number that is not rational is called irrational.

Proof by Contradiction

First assume that a proposition is not true, then through that premise and logic find a contradiction that shows our original premise must have been incorrect, and therefore, the proposition was true.

For one proposition p , assume $\neg p$ is true, then find a contradiction that shows $\neg p$ is false, so p is true.

For an implication, $p \rightarrow q$, assume p and $\neg q$ are true, then find a contradiction that shows either $p \rightarrow q$ or $\neg q \rightarrow \neg p$.

Proving Conditional Statements by Contradiction

Example

Prove that “If $n^3 + 5$ is odd, then n is even.”

Solution

Let p : If $n^3 + 5$ is odd, and q : n is even.

Assume p and $\neg q$, i.e. $n^3 + 5$ is odd, $n \in \mathbb{Z}$, and n is odd. $\therefore n = 2k + 1$ for $k \in \mathbb{Z}$

We show that the above assumption leads to a contradiction.

$$\begin{aligned}n^3 + 5 &= (2k + 1)^3 + 5 \\&= 8k^3 + 12k^2 + 6k + 1 + 5 \\&= 8k^3 + 12k^2 + 6k + 6 \\&= 2(4k^3 + 6k^2 + 3k + 3) \\&= 2t \qquad t = 4k^3 + 6k^2 + 3k + 3, t \in \mathbb{Z}\end{aligned}$$

Therefore $n^3 + 5 = 2t$, i.e. $n^3 + 5$ is even. This contradicts the statement that $n^3 + 5$ is odd.

Proof by Contradiction with One Proposition

Example Proof that $\sqrt{2}$ is irrational.

Solution

Let p be the proposition “ $\sqrt{2}$ is irrational.”

We suppose that $\neg p$ is true.

$\neg p$ is the statement “It is not the case that $\sqrt{2}$ is irrational, i.e. $\sqrt{2}$ is rational.

We show that assuming $\neg p$ is true leads to a contradiction.

Assume $\sqrt{2}$ is rational.

Then, there exists 2 integers a and b such that $\sqrt{2} = \frac{a}{b}$, $b \neq 0$, and a and b have no common factors.

$$\sqrt{2} = \frac{a}{b}$$

$$2 = \frac{a^2}{b^2}$$

$$2b^2 = a^2$$

Cont.

Therefore a must be even. So a can be written as $a = 2c$, for some $c \in \mathbb{Z}$

$$2b^2 = (2c)^2$$

$$2b^2 = 4c^2$$

$$b^2 = 2c^2$$

Therefore b must be even.

Since a and b are both even, they have a common factor.

Therefore $\sqrt{2}$ is irrational.

Example

Proof by contradiction “If n is an integer and $3n + 2$ is even, then n is even.”

Solution

Let p : $3n + 2$ is even, and q : n is even.

Assume p and $\neg q$, i.e. $3n + 2$ is even, $n \in \mathbb{Z}$, and n is odd.

We suppose that $\neg q$ is true.

Since $3n + 2$ is even, then $3n$ is even by subtraction of 2 even integers. Then $3n - n$ must be odd by subtraction of an even and odd integer. However, $3n - n = 2n$, which is even by definition of an even number, which is a contradiction of our original assumptions.

$\therefore$ “If n is an integer and $3n + 2$ is even, then n is even.”

Proof by Cases

Used to prove a theorem by considering different cases separately.

The validity of a proof by cases rests on the equivalence:

$$[(p_1 \vee p_2 \vee \dots \vee p_n) \rightarrow q] \leftrightarrow [(p_1 \rightarrow q) \vee \dots (p_n \rightarrow q)].$$

The result must be true for all cases. If it fails for one case, then the result is false.

Typical cases

Case 1: n is even

Case 2: n is odd

Case 1: $n > 0$

Case 2: $n = 0$

Case 3: $n < 0$

e.g. Prove that if $n \in \mathbb{Z}$, then $n^2 + 3n + 5$ is an odd integer.

Proof We proceed by cases, according to whether n is even or odd.

case 1, n is even, then $n = 2k$, $k \in \mathbb{Z}$

So,

$$\begin{aligned}n^2 + 3n + 5 &= 4k^2 + 6k + 5 \\&= 2(2k^2 + 3k + 2) + 1 \\&= 2r + 1 \qquad r = 2k^2 + 3k + 2, r \in \mathbb{Z}\end{aligned}$$

case 2, n is odd, then $n = 2k + 1$, $k \in \mathbb{Z}$

$$\begin{aligned}n^2 + 3n + 5 &= 4k^2 + 4k + 1 + 6k + 3 + 5 \\&= 2(2k^2 + 5k + 4) + 1 \\&= 2r + 1 \qquad r = 2k^2 + 5k + 4, r \in \mathbb{Z}\end{aligned}$$

Thus, $n^2 + 3n + 5$ is an odd integer.

PROVING NON-CONDITIONAL STATEMENTS

Biconditionals

In order to establish the truth of the statement $p \leftrightarrow q$, use the fact that $(p \leftrightarrow q)$ is equivalent to $(p \rightarrow q) \wedge (q \rightarrow p)$ and prove both implications using any of the methods of proof.

The validity of this approach is based on the tautology

$$(p \leftrightarrow q) \leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)$$

Example. Prove the theorem that if n is an integer, then n is odd if and only if n^2 is odd.

Proof

In order to prove this statement, we must prove two implications:

- If n is odd, then n^2 is odd.
- If n^2 is odd, then n is odd.

Proof of (a): We give a direct proof of this statement. Assume n is an odd integer. Then $n = 2k + 1$ for some integer k .

Then $n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$, which is twice an integer plus 1. Thus, n^2 is odd.

Proof of (b): We give a proof of the contrapositive of this statement:

“If n is even (not odd), then n^2 is even (not odd)”.

Assume n is an even integer.

Then $n = 2k$ for some integer k .

Then $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$, which is an even integer.

∴ Proven.

The End